# Problem Statement :- 9

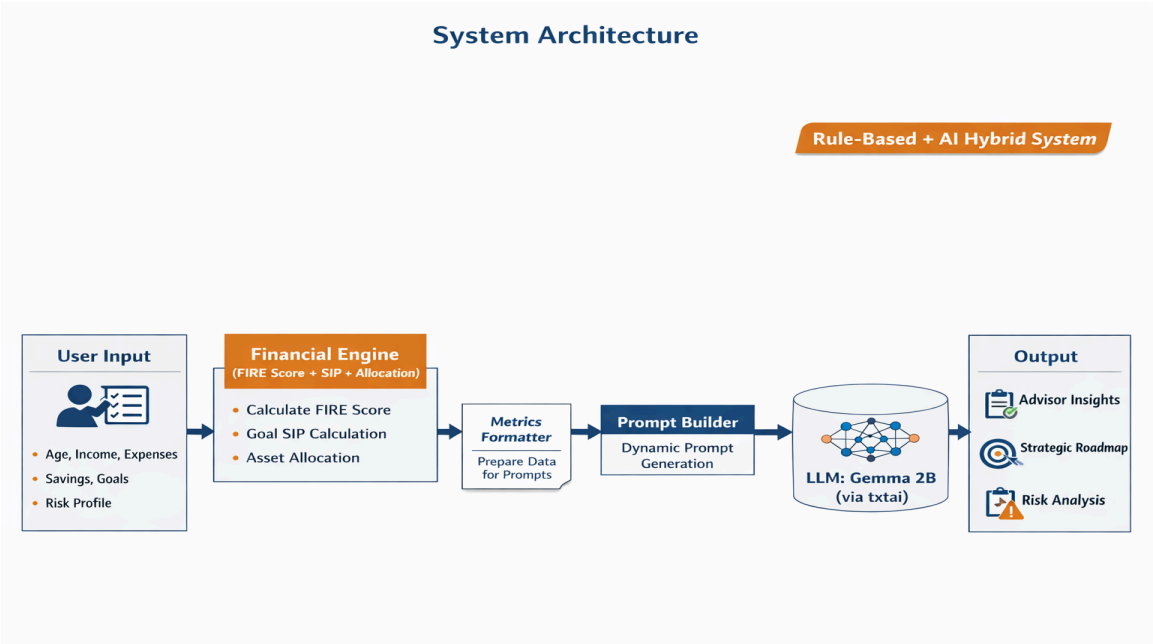
*95% of Indians don't have a financial plan. Financial advisors charge ₹25,000+ per year and serve only High Net-worth Individuals (HNIs). Build an AI-powered personal finance mentor that lives inside ET, turns confused savers into confident investors, and makes financial planning as accessible as checking WhatsApp*

- **WHAT YOU MAY BUILD**

1. ***Financial Independence, Retire Early (FIRE) Path Planner —*** *User inputs age, income, expenses, existing investments, and life goals. AI builds a complete, month-by-month financial roadmap: Systematic Investment Plan (SIP) amounts per goal, asset allocation shifts, insurance gaps, tax-saving moves, and emergency fund targets.*

***Prototype built-up by :- AI Devilops***
*Team Lead and member - Deepak Kaura*

## Prototype's Architecture :




## How the Input and Output works (Step-wise Flow)

*1. User provides financial inputs such as age, income, expenses, investments, and goals.*
*2. System calculates key metrics:*
*- Savings rate*
*- Target FIRE corpus (25× annual expenses)*
*- Investment shortfall*
*3. A FIRE score is computed based on savings rate, returns, inflation, and shortfall.*
*4. The score is mapped to a category: Critical, Weak, Moderate, Strong, or Optimized.*
*5. Based on the category, a predefined prompt template is selected.*
*6. User financial data is formatted into structured inputs.*
*7. The prompt is passed to the LLM (Gemma 2B via txtai).*
*8. LLM generates:*
*- Personalized financial advice*
*- Strategic roadmap*
*- Risk assessment*
*9. System also calculates:*
*- Goal-based SIP recommendations*
*- Suggested asset allocation (Equity vs Debt)*
*10. Final output is displayed to the user.*

## Test Scenarios

*Scenario 1: Low Savings User*
*- Input: Low income, high expenses, minimal investments*
*- Expected Output:*
  *- Low FIRE score*
  *- Category: Critical or Weak*
  *- Advice focused on expense reduction and income growth*

*Scenario 2: Balanced User*
*- Input: Moderate savings rate and stable investments*
*- Expected Output:*
  *- Moderate FIRE score*
  *- Category: Moderate*
  *- Advice focused on optimization and consistency*

*Scenario 3: High Savings User*
*- Input: High income, high savings, strong portfolio*
*- Expected Output:*
  *- High FIRE score*
  *- Category: Strong or Optimized*
  *- Advice focused on wealth preservation and early retirement*

*Scenario 4: Goal Heavy User*
*- Input: Multiple large financial goals*
*- Expected Output:*
  *- Higher SIP recommendations*
  *- Detailed goal planning output*

## Tech Stack ***(Prerequisite…libraries to run the prototype)***

*- Python (Programming language)*
*- LangChain (Prompt orchestration)*
*- txtai (LLM inference engine)*
*- Gemma 2B (Language model)*
*- Pandas (Data processing)*
*- Gradio (Deployment)*

## Limitations

*- Assumes constant income, expenses, and savings behavior*
*- Uses fixed return and inflation assumptions*
*- LLM responses may vary slightly due to generative nature*
*- Simplified retirement model (does not include dynamic withdrawal strategies)*

## Future Improvements

*- Implement time-series based simulations to evaluate FIRE probability under different market conditions, including visual insights and trend analysis.*

*- Enhance prompt engineering and upgrade to more advanced LLM/GenAI models for improved accuracy, personalization, and financial reasoning.*

*- Develop personalized investment optimization strategies based on user risk profile, goals, and market assumptions.*

*- Integration with real-time financial and market data APIs for dynamic and up-to-date analysis.*

*- Development of an interactive user interface (e.g., web dashboard) for better user experience and visualization.*

*- Scenario-based planning (e.g., job loss, salary growth, inflation spikes) to test financial resilience.*

*- Automated portfolio tracking and rebalancing recommendations.*

*- Multi-user profile support with comparative financial insights.*

*- Explainable AI (XAI) layer to justify recommendations and improve user trust.*

*- Mobile-friendly or app-based deployment for wider accessibility. (like ET's Whatsapp feature)*